Structure over material: mesocosm experiments reveal shelter design principles for threatened freshwater crayfish restoration

Olivier V. Raven   Robin Holmes   Ian A. K. Kusabs
Francis J. Burdon   Deniz Özkundakci

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Freshwater crayfish are among the most threatened invertebrates globally, with habitat loss a primary driver of population decline, yet the optimal design of habitat enhancement structures for their restoration remains largely unknown. Four controlled mesocosm experiments with the endemic Aotearoa-New Zealand freshwater crayfish kōura (*Paranephrops planifrons*) examined how shelter structure, material composition, body size, and group dynamics influence refuge selection. Individual kōura strongly preferred a wooden log with enclosed drilled holes, but when housed in mixed-size groups, kōura preference shifted to flat and piled stone configurations, providing more complex refugia for different-size individuals simultaneously. Flat stone structures were preferred over wood splits and this was strongest for large individuals. When structural dimensions were held constant, no significant preference was detected between concrete and wooden brick structures, indicating that material composition alone does not drive refuge selection. Size class effects on shelter preference were quantified using Bayesian categorical mixed-effects models, which provide an appropriate framework for multi-choice repeated-measures designs with categorical outcomes and nested group structures. These mesocosm findings suggest that stone-based configurations warrant field evaluation as habitat enhancement structures for kōura restoration, given their structural complexity across size classes and alignment with the cultural values associated with restoring this taonga species.

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1 Introduction

Crayfish are among the most threatened freshwater invertebrates globally, with 32% of species assessed as threatened with extinction due to habitat loss, water quality decline, and invasive species (Richman et al., 2015). As keystone species and ecosystem engineers, crayfish play crucial roles in freshwater food webs through shredding of leaf litter, bioturbation of sediments, and nutrient cycling, and therefore their loss can have profound cascading effects on freshwater ecosystems (Collier et al., 1997; Momot, 1995; Parkyn et al., 2001). The increasing degradation of freshwater habitats therefore not only threatens crayfish populations, but also risks disrupting the ecosystem processes crayfish sustain.

Habitat complexity is a key driver of crayfish population persistence, as structurally diverse habitats provide physical refugia from predators, foraging opportunities, and shelter during moulting when individuals are particularly vulnerable (Hammond et al., 2006). Shelter availability directly increases crayfish survival and growth under predation pressure (Adams, 2007), making refuge quality a critical consideration for both natural habitat persistence and restoration design. Crayfish use a wide variety of natural and artificial structures as shelter, including rocks, woody debris, and even anthropogenic objects such as bottles and discarded appliances (Adams, 2014; Parkyn et al., 2009), demonstrating considerable flexibility in habitat use. In the context of freshwater restoration, increasing structural complexity through providing shelters has improved crayfish abundance and restocking success in degraded streams and lakes (Huolila et al., 1997; Johnsen & Taugbøl, 2008), and the availability of coarser substrate has also been shown to improve juvenile survival rates (Mazlum et al., 2017). These findings indicate that enhancing habitat complexity can be a useful management tool for threatened crayfish populations. While artificial aquatic habitats have received attention in marine systems (Frehse et al., 2025), equivalent systematic comparisons of shelter configurations for freshwater crayfish restoration remain largely absent from the literature.

Shelter selection in crayfish is strongly influenced by body size, as individuals actively select refugia proportional to their dimensions. Crayfish individuals selected stones more than 3 times longer and 1.25 times wider than their carapace length, with larger males consistently selecting larger stones (Streissl & Hödl, 2002). Body size and shelter dimensions have direct competitive consequences as crayfish in undersized shelters initiated and won more fights than those in appropriately sized shelters, suggesting that shelter fit influences an individual's assessment of its own fighting ability (Percival & Moore, 2010). Social dynamics further complicate shelter use when refuge availability is limited, as competitive interactions among conspecifics determine which individuals gain access to preferred shelters. Dominance hierarchies based on body size and reproductive status show that males and maternal females often are able to overtake shelters from non-maternal females, while adults consistently outcompete juveniles for shelter access (Figler et al., 2005). Body weight ratio between contestants is a strong predictor of shelter takeover success, although prior ownership can offset size disadvantages (Ranta & Lindström, 1993), particularly when both individuals hold high-quality shelters (Takahashi et al., 2019). At the population level, habitat complexity buffers these competitive effects, with juveniles in structurally complex habitats showing higher survival, faster growth, and greater moulting rates as more shelter availability reduces agonistic interactions (Olsson & Nyström, 2009). Together, shelter use in crayfish reflects an interaction between individual body size and social competitive dynamics, suggesting that effective habitat enhancement structures must provide a range of shelter sizes to support different life stages and size classes simultaneously.

The material and structural properties of shelters influence their value as refugia. Studies comparing shelter configurations show that the availability and size of enclosed spaces matters more than total structure volume on reef fish assemblages (Hylkema et al., 2020). Additionally, enclosed structures such as PVC pipes also provide greater protection against predatory fish than stone or vegetation structures, likely due to their consistent hole dimensions (Zhao et al., 2024). The use of synthetic materials like PVC pipes in restoration ecology brings ecological concerns, particularly in contexts where introducing non-native materials may conflict with conservation values or indigenous cultural frameworks (Cooke et al., 2023; Wehi & Lord, 2017). Natural materials such as stones and wood offer ecologically compatible alternatives, yet the importance of structural configuration versus material composition for crayfish shelter preference remains largely unknown.

In Aotearoa-New Zealand (hereafter Aotearoa), the endemic freshwater crayfish kōura (*Paranephrops planifrons*) faces conservation pressure from habitat loss, water quality degradation, and predation by invasive species (Kusabs et al., inpress; Lee et al., 2025). Kōura are taonga (culturally treasured) species for Māori, the indigenous people of Aotearoa, and hold significant ecological and cultural value in freshwater ecosystems (Kusabs & Quinn, 2009). In their natural habitat, kōura are positively associated with overhanging banks, coarse substrates, and woody debris (Jowett et al., 2008; Parkyn et al., 2009; Raven et al., inreview; Usio & Townsend, 2000). Kōura have been shown to significantly increase refuge use in the presence of predators (Raven et al., indraft; Shave et al., 1994). However, a systematic study on shelter design preference on crayfish remains largely absent from the literature. Habitat restoration by increasing shelter availability represents a promising management tool for kōura recovery, and the use of natural materials aligns with ecological and cultural values set for these naturally and culturally important lakes.

In this study four controlled mesocosm experiments were conducted to identify kōura preferences among different structures varying in material and configuration, to examine how individual behaviour, body size, and group interactions influence refuge selection. We predicted that 1) Structurally complex structures providing enclosed crevices would be preferred over isolated or open structures, as enclosed spaces reduce predation risk. 2) Structure preference would be size-dependent, with larger kōura selecting structures with larger refuge spaces (Percival & Moore, 2010; Streissl & Hödl, 2002). 3) Stone-based structures would be preferred over wood due to greater structural stability, and 4) material type alone would not influence refuge preference between concrete and wooden bricks. The results provide practical guidance for designing cost-effective, ecologically appropriate habitat enhancement structures to support kōura population recovery.

2 Materials and Methods

2.1 Animal collection and tank setup

In mid-August 2025, 72 kōura were sourced from lakes Ōkāreka and Tikitapu, located within the Rotorua Te Arawa Lakes area in the North Island of Aotearoa. Collected animals were transported to the Aquatic Research Centre at the University of Waikato in Hamilton. Individuals were assigned to one of three size classes based on orbital carapace length (OCL): small ($<$ 22 mm), medium (22–30 mm), and large ($>$ 30 mm). Each animal was marked with a unique identification number written on the carapace using an orange oil-based paint marker (Sharpie, medium point) (Ramalho et al., 2010).

Kōura were acclimatised for 12 days in outdoor concrete holding tanks (Ø 1.4 m, 0.5 m deep), each filled to a depth of 0.4 m with dechlorinated tap water conditioned for three months.

Paving sand was used as substrate, and shelters were provided as PVC pipes, broken terracotta pots, and perforated building bricks. Each holding tank housed 10–12 kōura. Kōura were fed every second day with commercial carnivore pellets (Hikari Tropical, Kyorin Food Industries Ltd., Japan; animal protein 47%).

Experiments were conducted outside in six circular concrete tanks (1425 L; Ø 1.85 m, 0.53 m deep), each filled to a depth of 0.4 m with dechlorinated tap water conditioned for three months. Paving sand was used as substrate and tanks were closed off with a lid constructed of two layers of shade cloth to prevent escape and minimise solar heating. Fresh dechlorinated tap water and an air stone were added to each tank in between the experimental rounds. To limit thermal stress, experiments were conducted in spring (September to October). Because kōura are a taonga (treasured) species, permission was obtained from Te Arawa Lakes Trust and Te Komiti Whakahaere. All experimental procedures were conducted under University of Waikato animal ethics committee approval (protocol number: 1207; see Ethical Approval section).

2.2 Exp 1: Single kōura

This experiment assessed individual kōura structure preference by offering single animals a choice among four structures (Figure 1). The structure-types were: SingleStone (ten quarried stones (150–250 mm) placed separately), FlatStone (ten quarried stones arranged tightly without stacking), StonePile (ten quarried stones stacked to form a pile), and WoodLog (a tānekaha (*Phyllocladus trichomanoides*) log (500 mm long, 150 mm diameter) containing six drilled holes (70 mm deep, 32 mm diameter)). Six experimental tanks were used, each housing a single medium-sized kōura. In total, 30 kōura consisting of 15 females with mean OCL and weight $\pm$ SD: 26.9 $\pm$ 2 mm, range: 23.2–29.8 mm, 12.5 $\pm$ 3.3 g and 15 males: 26.1 $\pm$ 2.2 mm, range: 22.6–30 mm, 11.9 $\pm$ 3.1 g were tested across five experimental replicates, with each replicate lasting 22 hours.

Each experimental tank was divided into four quadrants, with one structure type placed at random (randomizer.org) in each quadrant. If a kōura was not using a structure, TankWall was noted as its location which represents the wall of the tank. For each trial, a single kōura was placed inside a vertically oriented plastic tube (50 mm long, open at one end and sealed at the other) positioned at the centre of the tank, with the sealed end resting on the substrate. This ensured a standardised starting position from which the kōura could exit in any direction. The time taken for the kōura to locate its first structure was recorded (L0). On the following day, the final location of the kōura was noted (L1) after which it was removed from the tank. Tanks were reset and structure placements re-randomised prior to the next trial. Each kōura was used in a single trial only and was not reused across replicates. To confirm nocturnal activity, three GoPros (Hero 3) were placed in one tank at the end of the day and recorded for five hours.

2.3 Exp 2: Group kōura

Building on experiment 1, this experiment examined how group dynamics and body size influenced structure preference when kōura were housed together. The six experimental tanks were used and configured with the same structure types and randomisation procedure as in experiment 1 (Figure 1). In contrast to the single-animal trials, each tank housed a group of six male kōura, comprising two small, two medium, and two large individuals. Size classes were defined by OCL: small (mean OCL and weight $\pm$ SD: 19.2 $\pm$ 2 mm, range: 15.8–21.6 mm, 3.9 $\pm$ 2.1 g), medium (26.4 $\pm$ 1.6 mm, 23.5–29.5 mm, 11.6 $\pm$ 2.4 g), and large (36.3 $\pm$ 4.2 mm, 30.7–43.3 mm, 31.8 $\pm$ 11.2 g). Each group was introduced to the centre of the tank within an open-top plastic container (150 $\times$ 150 $\times$ 85 mm). The container was removed, and the initial location of each individual was recorded. On the following day, the final location of each kōura was documented, before all individuals were removed from the tank. This procedure was repeated across five experimental replicates reusing the same six groups of animals.

2.4 Exp 3: Wood vs Stone

Using the same groups from experiment 2, this experiment narrowed the choice to two structure types, stone and wood, to directly compare preference between these materials. The structure types were: (1) a flat stone structure (FlatStone) constructed from twelve stones placed tightly together, and (2) a wood structure consisting of twelve wood splits (WoodSplit) made from the tānekaha logs (Figure 1). Each group was released at the centre of the tank, and the initial structure selected by each kōura was recorded. The following day, the final location of each kōura was documented. This procedure was repeated three times, reusing the same six groups of animals.

2.5 Exp 4: Bricks

This experiment tested whether kōura showed a preference between concrete and wooden perforated brick structures, to isolate the effect of material on structure preference. Two perforated building bricks (BrickConcrete) of 80 $\times$ 80 $\times$ 225 mm with five holes (Ø 30 mm) were tested against two wooden bricks (BrickWood) made from Western red cedar (*Thuja plicata*), constructed to match the building bricks in external dimensions and hole size. To prevent floating, a solid clay brick of identical dimensions was placed on top of each wooden brick (Figure 1). New groups of ten kōura and four experiment tanks were used for this experiment, comprising a total of 40 individuals consisting of 18 females with mean OCL and weight $\pm$ SD: 26.2 $\pm$ 3.1 mm, range: 20.2–30.8 mm, 12.2 $\pm$ 4.2 g and 22 males: 26 $\pm$ 2.1 mm, range: 20.9–29.5 mm, 11.6 $\pm$ 3.2 g. Due to reduced animal availability, individuals were drawn from a slightly broader size range than medium-size individuals, resulting in unbalanced size class representation (small: n = 4, medium: n = 35, large: n = 1). In each of the four experimental tank quadrants one brick was placed, with similar brick types positioned opposite to one another. Each group of kōura was released at the centre of the tank and the following day, the final location of each kōura was documented. This experiment was repeated four times by reusing the four groups of ten kōura.

2.6 Data analysis

To test the difference between the four structure types selected the next day by the single kōura, a multinomial goodness-of-fit test was performed using the XNomial package (Engels, 2014). TankWall was excluded from preference tests as it represents the absence of structure choice rather than a structure type preference. Additionally, pairwise likelihood-ratio G-tests with Bonferroni correction were used to identify which structure types differed significantly from each other. To test whether structure type use differed between sexes, Fisher's exact test with simulated p-values (9999 replicates) was used due to small expected counts. A permutation-based symmetry test was applied to assess whether structure type transitions between L0 and L1 were directional rather than random.

To test whether overall structure preference existed at the group level, one-sample Wilcoxon signed-rank tests were applied to group-round proportions for each structure type, testing against a null expectation of 0.25 (equal distribution among four structure types). P-values were Bonferroni adjusted for multiple comparisons. Normal approximation was used due to ties arising from discrete group counts. Bayesian categorical mixed-effects models were used to test whether body size or sex affected structure preference across experiments 2, 3 and 4. The outcome is categorical and individual animals within group within round create a three-level nested random effect structure. Standard frequentist mixed-effect approaches such as lme4 (Bates et al., 2015) do not support categorical outcomes with complex nested random effects structures, making Bayesian estimation the appropriate choice for this design (Bolker et al., 2009; McCarthy, 2007). Model convergence was assessed using the Gelman-Rubin diagnostic (all $\hat{R} = 1.00$) and divergent transition counts (all experiments: 0 divergences), confirming reliable posterior sampling (Gelman & Rubin, 1992). To test whether size classes affected structure preference while accounting for repeated measurement of individuals and groups, a Bayesian categorical mixed-effects model was fitted using the brms package (Bürkner et al., 2025), with structure type as the outcome, size class as a fixed effect, and group identity, individual identity nested within group, and experimental round as random intercepts. The Stuart-Maxwell test was used to test for statistical differences between stage L0 and L1.

To test the difference between FlatStone and WoodSplit the Wilcoxon signed-rank test was applied to group-round proportions for each structure type, testing against a null expectation of 0.5. P-values were Bonferroni adjusted. To test whether size classes affected structure preference while accounting for repeated measurement of individuals and groups, a Bayesian categorical mixed-effects model was fitted with size class as a fixed effect, following the same structure as described for experiment 2. The Stuart-Maxwell test was used to test for statistical differences between stage L0 and L1.

To test whether kōura showed a preference between BrickConcrete and BrickWood, the Wilcoxon signed-rank test was applied to group-round proportions for each brick type, testing against a null expectation of 0.5 and p-values were Bonferroni adjusted. A paired Wilcoxon signed-rank test was used to directly compare group-round proportions between BrickConcrete and BrickWood. As size class was unbalanced, the effect of sex on brick preference was tested using a similar Bayesian categorical mixed-effects model was fitted with sex as a fixed effect, following the same structure as described for experiment 2. Finally, as only next-day locations (L1) were recorded in this experiment, no directional shift analysis was performed.

3 Results

3.1 Exp 1: Single kōura

Individual kōura showed a strong preference for WoodLog, selecting it significantly more often than SingleStone, FlatStone, and StonePile (pairwise G-test, Bonferroni adjusted: p = <0.001, 0.011, 0.004, respectively; overall multinomial test: p = <0.001; Figure 2). No significant differences were found among the stone structure types. Overall, WoodLog was selected by 67% of kōura at L1 (males: 80%; females: 53%), and no kōura were found at TankWall, indicating all individuals actively selected a structure. Structure preference did not differ significantly between sexes (Fisher's exact test: $p = 0.450$). The symmetry test indicated that structure type use shifted between L0 and L1 ($p = 0.002$), suggesting kōura actively relocated to preferred structure types overnight rather than remaining at their initial location. The GoPro footage also shows kōura emerging, walking and climbing on the structures after sunset.

3.2 Exp 2: Group kōura

When housed in mixed-size groups, kōura showed a significant overall preference for stone-based structures (Figure 3). FlatStone was used significantly more ($p = 0.004$), while SingleStone (p = <0.001) and WoodLog ($p = 0.039$) were significantly avoided. StonePile did not differ from random expectation ($p = 1.000$), and did not differ significantly from FlatStone ($p = 0.117$). No kōura were found at TankWall, indicating active structure selection across the group. Size classes had limited but credible effect on structure preference. Medium sized kōura showed a weaker preference for FlatStone over SingleStone compared to small kōura (= -1.75, 95% CI: -3.66 to -0.09; posterior probability = 0.98), while no other size class effect was credible. Overall, FlatStone remained the most selected structure type across all size classes, but its strength of preference declined from small (posterior probability of top choice = 85%) to medium (53%), and with large kōura showing similar preference between FlatStone and StonePile (41% vs 38%). Structure type use shifted directionally between L0 and L1 (Stuart-Maxwell test: p = 0.001), indicating that kōura actively relocated to preferred structure types overnight.

3.3 Exp 3: Wood vs Stone

Kōura consistently selected FlatStone over WoodSplit, with the preference growing with body size (Figure 4). At the group level, FlatStone was used significantly more than expected ($p = 0.001$), while WoodSplit was significantly avoided ($p = 0.004$). Only 1% of kōura were found at TankWall, suggesting almost all kōura selected one of the two structure types. The Bayesian model indicated that preference for FlatStone strengthened with body size, with the posterior probability of FlatStone being the top-ranked structure increasing from small (62%) to medium (78%), to large kōura (96%). Large kōura showed a particularly strong and credible preference for FlatStone over WoodSplit (= -2.4, 95% CI: -4.39 to -0.8; posterior probability = > 0.99), while medium kōura did not differ from small kōura. Structure type use shifted directionally between L0 and L1 (Stuart-Maxwell test: $p = 0.003$), with kōura relocating toward FlatStone overnight.

3.4 Exp 4: Bricks

Kōura used both brick types, showing no significant preference between BrickConcrete and BrickWood ($p = 0.137$; Figure 5), with mean group proportions of 47% and 53% respectively. Neither brick type differed significantly from equal distribution ($p = 0.273$). 9% of kōura were found at TankWall, which is higher than in any previous experiment. Sex had no credible effect on preference between brick types (= -0.18, 95% CI: -1.29 to 0.85; posterior probability = 0.64), indicating that concrete and wooden perforated brick structures of identical dimensions were both suitable refuge for male and female kōura.

4 Discussion

4.1 Structure matters more than material

Across four experiments, refuge configuration and structural stability consistently predicted kōura habitat selection more strongly than material composition alone. Experiment 4 showed that when kōura were presented with similar refuge sizes in BrickConcrete and BrickWood no preference for material was detected. In experiment 3 FlatStone and WoodSplit were tested against each other and showed a difference in structure preference likely due to the stability of the stones compared to the wooden splits. The splits sank but created less structurally stable refuge places in the sandy substrate than the FlatStone structures [O.V. Raven, pers. obs.]. Experiment 2 showed that when a group of kōura was presented with multiple structure types, the structures that provide the most crevice complexity drove group-level preference. This was the case for FlatStone followed by StonePile for harbouring the most kōura. In experiment 1 the preference was different as individual kōura preferred WoodLog as refuge, this was consistent with prediction 1, as single kōura probably searched the refuge that would give most protection

(Jurcak et al., 2016), and the drilled holes in the WoodLog offered most protection similar to what has been found for perforated brick and PVC tube shelters under catfish predation (Adams, 2007; Zhao et al., 2024). The directional shift in structure type use between L0 and L1 confirmed across all experiments that kōura actively relocated overnight rather than remaining at their initial location, supporting the interpretation that next-day locations reflect genuine refuge preference rather than chance displacement. The preference shift from WoodLog in single to stone structures in group experiments was unexpected. It might be explained by size mismatch between fixed hole dimensions and the range of body sizes present in each group, while the large surface area of FlatStone allowed multiple individuals to shelter simultaneously and reduce competitive displacement. Learning across rounds may also have contributed, as WoodLog was most selected structure in round 1 of experiment 2 (17 out of 36 times) but not in subsequent rounds, suggesting kōura may have shifted preference after initial exploration. This learning was possible as the groups of kōura in experiment 2 were reused all five rounds, while the single kōura in experiment 1 were only used once. The reuse of groups across rounds was necessary given the limited availability of kōura and ethical constraints on collection numbers. The number of independent groups was low across experiments (six in experiments 2 and 3, four in experiment 4), constraining the statistical power for detecting group-level effects (Hurlbert, 1984). Despite these constraints, consistent patterns in size-dependent preference were evident across experiments.

4.2 Body size and competitive dynamics

Size class consistently influenced refuge preference across experiments, but the pattern differed by context. In experiment 3, larger kōura showed a stronger preference for FlatStone over WoodSplit than smaller individuals, consistent with size dependent shelter selection documented in crayfish (Percival & Moore, 2010; Streissl & Hödl, 2002). This may reflect larger kōura being more capable of asserting dominance over preferred stable refugia through competitive displacement, which is in line with the size-based dominance hierarchy documented in kōura populations (Figler et al., 2005; Ranta & Lindström, 1993; Stewart & Tabak, 2011). In experiment 2, where four structure types were available, small kōura showed the strongest preference for FlatStone while large kōura distributed more evenly between FlatStone and StonePile, suggesting that FlatStone's large continuous surface area provides refuge spaces proportional to smaller body dimensions (Streissl & Hödl, 2002), while larger individuals find adequate crevice space in either stone configuration. The absence of sex effects in experiment 1 and 4 suggests that refuge preference in kōura is driven primarily by body size and competitive dominance rather than sex (**?**). When refuge availability is limited, competitive hierarchies displace smaller and recently moulted individuals from preferred refugia, leaving them exposed (Figler et al., 2005; Ranta & Lindström, 1993). Providing sufficient refuge spaces with a range of crevice size can reduce agonistic interactions and benefit the whole population (Olsson & Nyström, 2009), supporting complex structure configurations over structures with fixed hole dimensions.

4.3 Restoration implications

The present study demonstrates that structural configuration drives refuge preference more strongly than material composition, with stone-based configurations preferred by groups and no preference detected between materials when dimensions were equal. These findings suggest that stone-based habitat enhancement structures warrant field evaluation in lake conditions, given their structural complexity and crevice diversity across size classes. The broader restoration literature supports both the structural principle and the choice of natural materials. Structurally stable crevices are important for freshwater species like crayfish (Huolila et al., 1997; Streissl & Hödl, 2002), but also for marine environments, as reef fish, marine shrimp, and lobsters benefit from habitat enhancement structures (Dickson et al., 2023; Hylkema et al., 2020; Magnelia et al., 2008; Spanier, 1994). Structures made from stone and wood have been shown to outperform synthetic alternatives and bare substrates (Cooke et al., 2023; Dickson et al., 2023; Huolila et al., 1997; Magnelia et al., 2008), and do not contribute to plastic pollution in the environment (Cooke et al., 2023). Locally sourced natural materials also align with kaitiakitanga, which is the Māori principle of environmental guardianship and preserves the natural character of culturally significant lake environments (Clapcott et al., 2018). For in-lake deployment, existing habitat surveys suggest that kōura are associated with coarser substrate (Jowett et al., 2008; Kusabs et al., 2015; Olsson et al., 2006), and cobbles of 50–400 mm have been previously used in stream and lake restoration (Huolila et al., 1997; Johnsen & Taugbøl, 2008). FlatStones and StonePile both provided shelter spaces across size classes, but for in-lake deployment stone pile structures are the preferred candidates for field evaluation as due to their elevated structure, stone piles are less likely to be buried into soft substrates than flat configurations (Garron et al., 2024). With the threat from invasive predatory fish, like catfish preying on kōura (Kusabs et al., inpress), enhancing habitat complexity which provides enclosed crevices can be important for kōura as refuge availability directly influences predation survival (Adams, 2007; Zhao et al., 2024). While these mesocosm findings provide clear design guidance, field validation in lake environments remains an essential next step, as in-lake conditions including water movement, natural substrate, and predator pressure may modify kōura preferences in ways not captured under controlled conditions.

4.4 Conclusion

Across four mesocosm experiments, structural configuration and crevice complexity consistently predicted kōura shelter preference more strongly than material composition. The shift from individual preference for enclosed wooden log shelters to group preference for stone configurations shows that social context fundamentally changes what makes a shelter valuable. Size-dependent preferences further suggest that structures accommodating multiple body sizes simultaneously benefit whole populations rather than only dominant individuals. Stone-based structures using locally sourced materials are the most promising candidates for field evaluation, aligning with ecological and cultural values associated with restoring this taonga species. Field validation remains the essential next step for translating these findings into effective kōura habitat restoration practice.

5 Acknowledgements

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5.1 Authors' contributions

Study conception and design were undertaken by O.V.R., R.H. and F.J.B. Experiments were conducted by O.V.R. Data analysis was performed by O.V.R. Local ecological knowledge and kōura were supplied by I.A.K.K. Manuscript drafting was led by O.V.R. All authors contributed to manuscript revision, approved the results, and the final manuscript.

5.2 Ethical approval

This study received ethical approval by University of Waikato animal ethics committee, protocol number: 1207. Additional permission to conduct experiments with kōura was granted by Te Arawa Lakes Trust and Te Komiti Whakahaere.

5.3 Declarations

The corresponding author has declared that none of the authors has any competing interests.

5.4 Data availability statement

The code and derived data supporting the findings of this study are openly available on GitHub (https://github.com/OlivierRaven/Koura_Mesocosm)and a rendered version of the analysis is hosted at https://olivierraven.github.io/Koura_Mesocosm/. The primary raw dataset is archived on Zenodo (https://doi.org/……). Peer review documents, including the cover letter and reviewer responses, are available in the manuscript repository in the interest of open and transparent science.

6 Figures

Figure 1: Overhead photographs of experimental tank configurations for each of the four mesocosm experiments. Experiment 1 (Single kōura) and Experiment 2 (Group kōura): four shelter types, WoodLog (tānekaha log with drilled holes), FlatStone (stones arranged flat), StonePile (stones stacked), and SingleStone (individual stones placed separately). Experiment 3 (Wood vs Stone): WoodSplit shelter (tānekaha wood splits) and FlatStone shelter placed in opposing halves of the tank. Experiment 4 (Bricks): BrickConcrete (perforated concrete bricks) and BrickWood (wooden bricks of identical dimensions) placed in opposing quadrants of the tank.

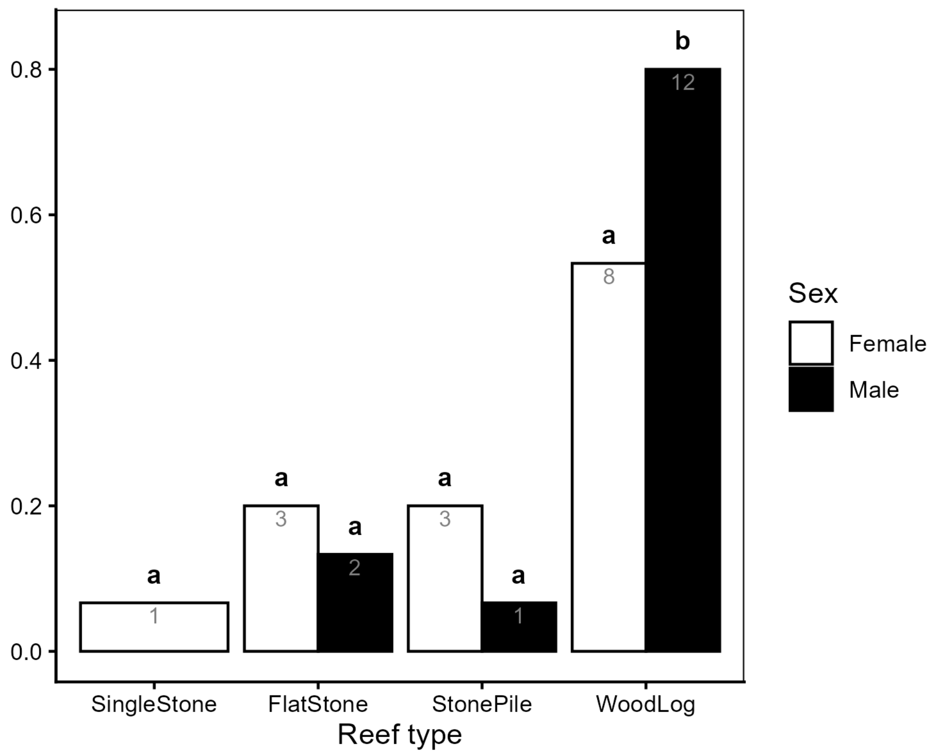

Figure 2: Proportion of individual kōura (Experiment 1) found at each reef type at final observation (L1), shown separately for female and male kōura. Bars show observed proportions with sample sizes (n) shown in white. Letters above bars indicate significant pairwise differences (G-test, Bonferroni adjusted).

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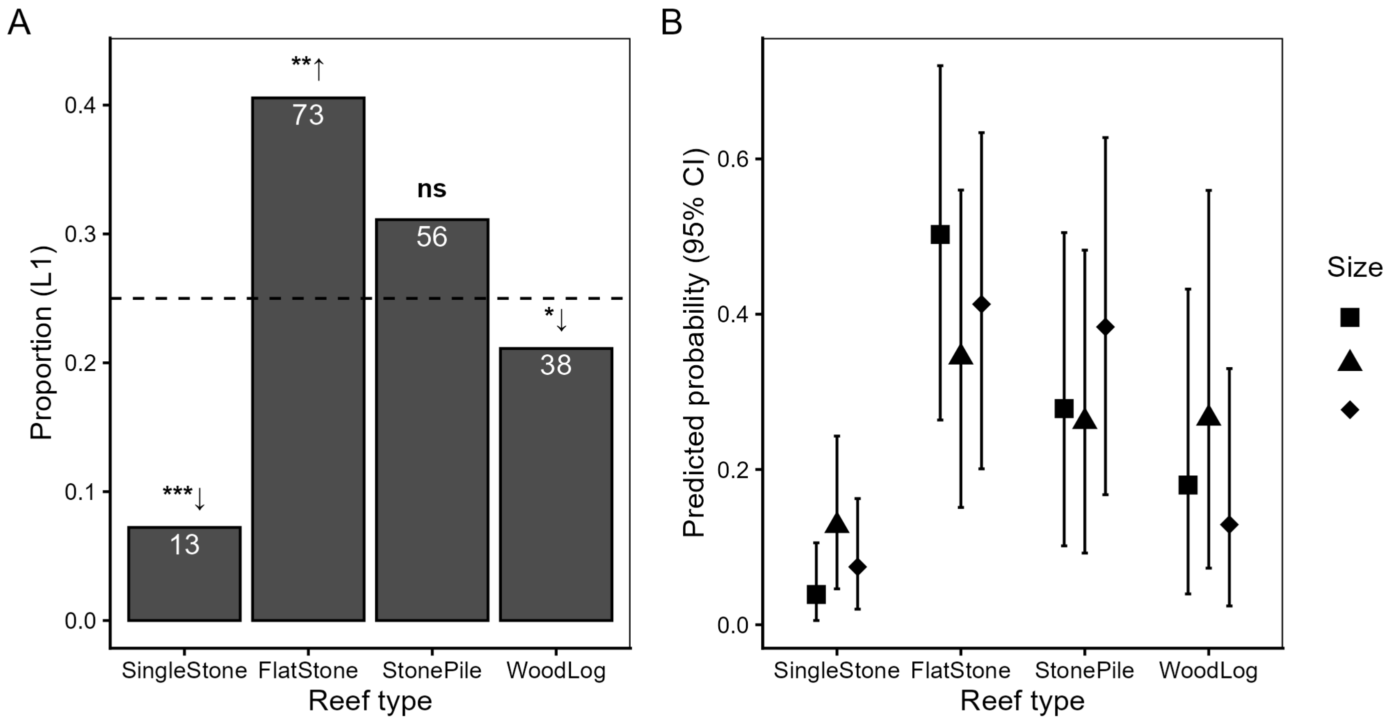

Figure 3: Reef type preference of group kōura (Experiment 2). (A) Proportion of kōura found at each reef type at L1 across all groups and rounds. The dashed line indicates the null expectation of equal distribution (0.25). Significance symbols indicate deviation from null expectation (Wilcoxon signed-rank test, Bonferroni adjusted): ↑ used more than expected, ↓ used less than expected. Sample sizes shown in white. (B) Posterior predicted probability of reef type use by size class from Bayesian categorical mixed model, with 95% credible intervals.

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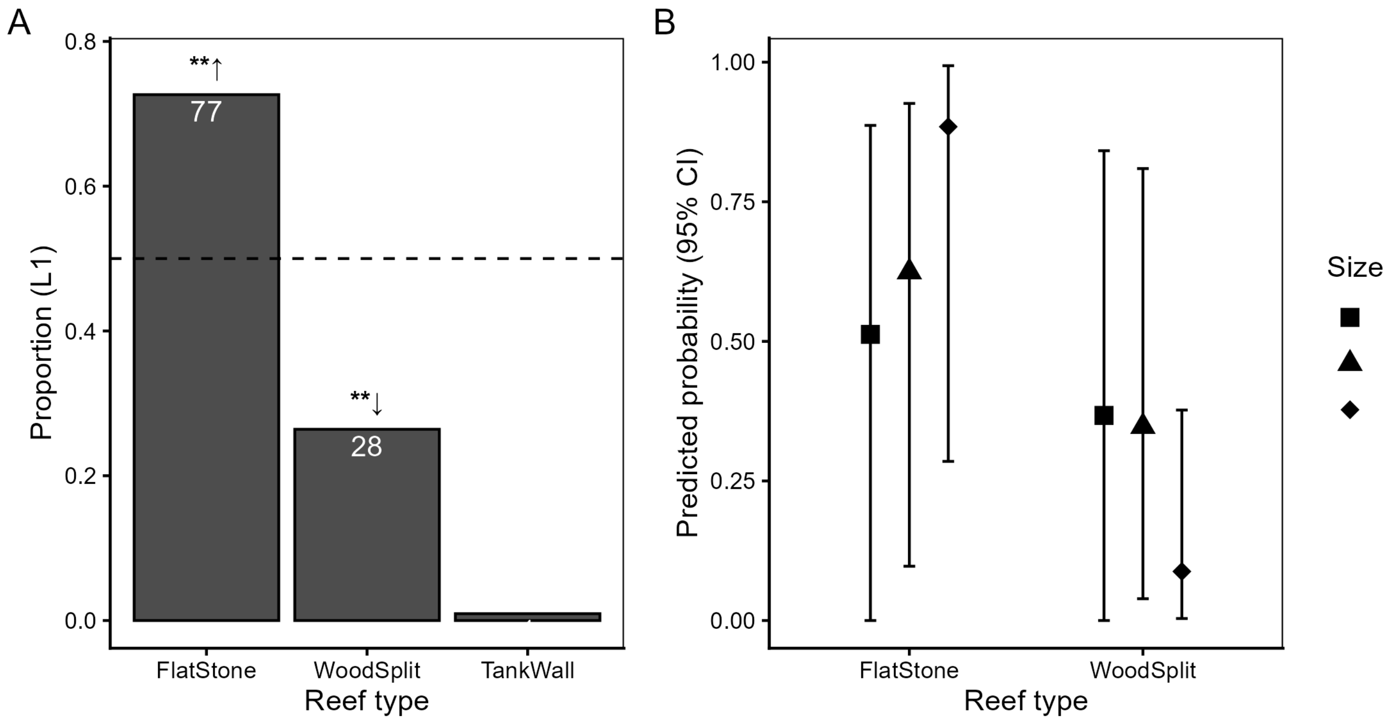

Figure 4: Reef type preference in stone versus wood comparison (Experiment 3). (A) Proportion of kōura found at each reef type at L1 across all groups and rounds. The dashed line indicates the null expectation of equal distribution (0.5). Significance symbols as in Figure 3. (B) Posterior predicted probability of reef type use by size class from Bayesian categorical mixed model, with 95% credible intervals. TankWall excluded from model predictions.

Source: Article Notebook

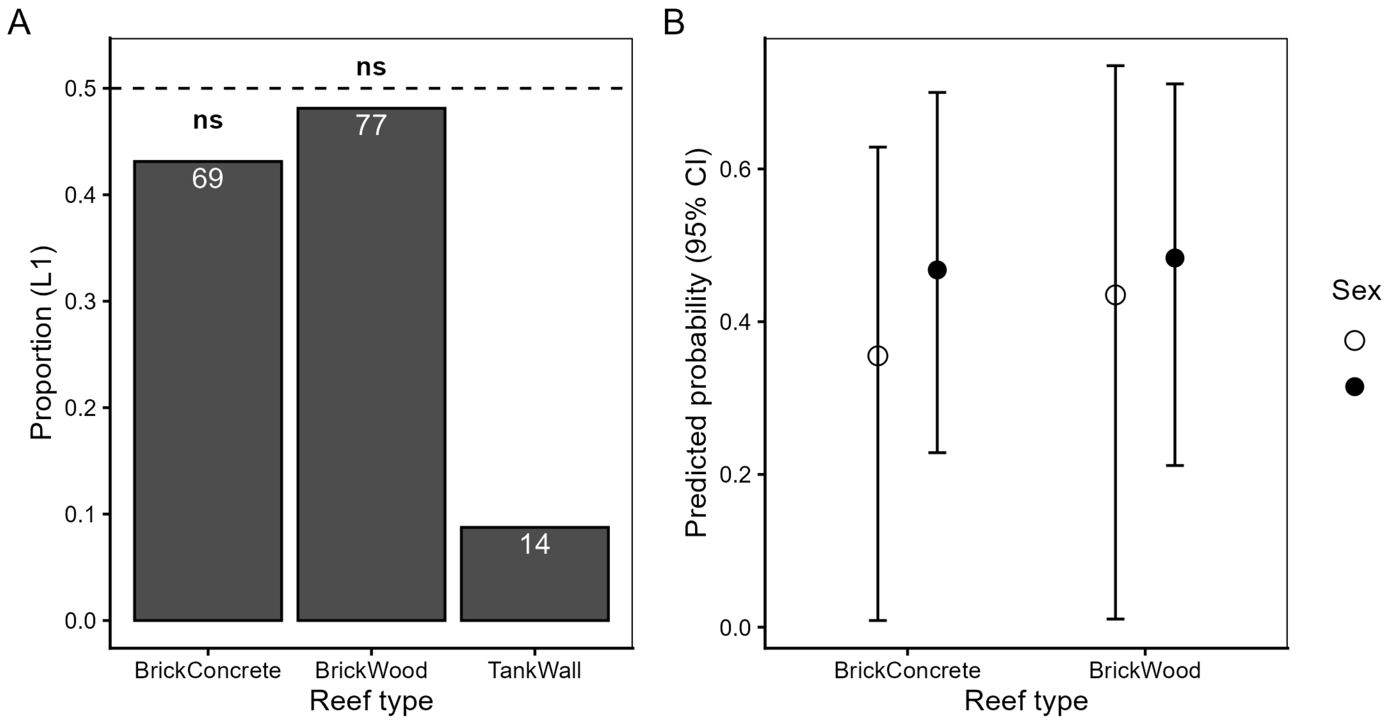

Figure 5: Brick material preference (Experiment 4). (A) Proportion of kōura found at each reef type at L1 across all groups and rounds. The dashed line indicates the null expectation of equal distribution (0.5). (B) Posterior predicted probability of reef type use by sex from Bayesian categorical mixed model, with 95% credible intervals. TankWall excluded from model predictions.

Source: Article Notebook

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